

Pushkal Pratap Singh

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SUMMARY

AI/ML and Backend Engineer building LLM gateways, RAG pipelines, and ML microservices with Python, LangChain, FastAPI, and vector databases. Smart India Hackathon 2025 National Finalist and AWS AI for Bharat 2nd Runner-Up.

EDUCATION

SRM Institute of Science and Technology, Ramapuram

B.Tech, Computer Science • CGPA: 9.06 / 10 (Sem 1–6)

Chennai, India

2023 – 2027

EXPERIENCE

Backend Developer Intern • Tranquix (Remote)

Dec 2025 – May 2026

- Developed and maintained REST API endpoints and database schemas supporting core product functionality in production.
- Partnered with frontend engineers on feature delivery, bug triage, and code review across sprint cycles.

AI Training Contractor • Outlier AI (Remote)

Jan 2025 – Dec 2025

- Evaluated LLM outputs across coding, reasoning, and math domains; authored prompt-response pairs and rubric-based assessments to improve model alignment.
- Performed prompt optimization and dataset quality audits, flagging hallucinations and ambiguous specifications across hundreds of task batches.

PROJECTS

Nyay Samiti • *Legal Document Intelligence Platform*

github.com/xpushkal/nyay-samiti

- Fine-tuned 7 transformer models (LegalBERT, RoBERTa, BART, T5, Sentence-BERT) on Indian legal corpora; achieved 90.4% F1 (NER), 84.16% accuracy (Risk Scoring), and 87.18% Exact Match (Q&A) with PyTorch and Hugging Face Transformers.
- Built a FastAPI ML microservice with a 7-stage sequential inference pipeline and spaCy preprocessing, delivering full legal document analysis in ~15s on CPU / ~3s on GPU.
- Orchestrated a 6-service production stack (FastAPI, MongoDB, Kafka, Zookeeper, Prometheus, Grafana) with Docker Compose for reproducible deployment and real-time model observability.

Prompt Intelligence Layer (PIL) • *Opinionated LLM gateway*

github.com/xpushkal/prompt-intel-layer

- Architected an OpenAI-compatible LLM gateway with intent-aware routing across OpenAI, Anthropic, Gemini, and local Ollama, exposing a unified `/v1/chat/completions` API with hot-reloadable YAML routing policies and prompt templates — async FastAPI, Postgres + pgvector, and Redis.
- Built a semantic cache (Redis + pgvector embeddings) and intent classifier that routes requests to the right model tier, cutting token cost and latency while preserving response quality; per-key budgets and circuit breakers prevent runaway spend.
- Shipped PII masking (Presidio + custom Indian-ID rules) with reversible de-tokenization, Pydantic-based structured output validation, and provider fallback chains; instrumented end-to-end with OpenTelemetry traces, Prometheus metrics, and a shadow-mode + replay endpoint for safe model rollouts against a golden eval set.

ZIT • *Git Workflow Automation Tool*

github.com/xpushkal/zit

- Shipped a serverless AI backend on AWS Lambda + API Gateway with a dual-model strategy (Claude 3 Sonnet for complex reasoning, Claude 3 Haiku for lightweight requests), cutting token cost while preserving response quality; deployed via AWS SAM/CloudFormation with scoped IAM and 27 pytest cases covering request flow, errors, and CORS.
- Implemented a RAG pipeline using FAISS for vector search and Amazon DynamoDB as the metadata store, powering grounded LLM context retrieval.

SKILLS

Languages: Python, SQL

ML / AI: LangChain, RAG Pipelines, Fine-tuning, Hugging Face Transformers, PyTorch, spaCy, LLM Gateways, Prompt Engineering

Vector Stores: Pinecone, FAISS, Chroma, pgvector

Backend: FastAPI, Pydantic, REST APIs, PostgreSQL, Redis, MongoDB

Cloud & DevOps: AWS (Lambda, API Gateway, DynamoDB, SAM, CloudWatch, Amplify, EC2), Azure, Docker, Docker Compose, Git, CI/CD

Observability: OpenTelemetry, Prometheus, Grafana

CS Foundations: Data Structures & Algorithms, Operating Systems, DBMS, System Design

ACHIEVEMENTS

- Smart India Hackathon 2025** — National Finalist (top teams from 50,000+ registrations).
- AI For Bharat 2026 (powered by AWS)** — 2nd Runner-Up (top teams from 96,000+ developers).